React refresher

Here is a quick refresher intended for readers who need a brief recap of the key React characteristics that we will rely on throughout our development process.

React is primarily a UI library that helps developers write interactive web applications powered by JavaScript. It includes

- *Components*—Reusable, self-contained pieces of UI (e.g., `ChatList`, `TextArea`).
- *JSX*—A syntax extension for JavaScript that allows writing HTML-like code in React. For example, instead of using `createElement( 'h1, 'Hello');` developers simply use `<hi>Hello</h1>`.
- *Props*—Read-only data passed from parent to child components.
- *State*—Mutable data that can change over time, managed within a component.
- *Hooks*—Functions that let you use state and other React features in functional components.
- *Virtual DOM*—React's efficient way of updating the UI by comparing virtual representations of the DOM.
- *Lifecycle methods*—Functions that run at different stages of a component's existence.

2.5 Building the major components

Now that we've outlined the goals and user experience, let's explore the project's structure, focusing on the key architectural components. We'll start with the frontend, where user interactions begin, and then move to the backend, which handles communication with our AI model.

2.5.1 Frontend

The frontend is where the UX comes to life. In this application, we use a component-based architecture to build fluid, responsive chat interactions.

The `<ChatPage/>` component serves as the central building block of the chat interface. Figure 2.2 shows how this component is rendered into the final page. This component is located in the src/pages folder and is responsible for rendering visual elements and managing user interactions. This component handles

- Displaying a list of past conversation messages
- Providing an input field for users to type their questions or prompts
- Submitting user input to the LLM for processing
- Displaying the LLM's generated response
- Incorporating features like automatic scrolling for long responses (as discussed earlier)

NOTE In a React application, the terms *page* and *component* are often used interchangeably, but they typically denote different organizational roles. A page (such as `ChatPage`, often found in a src/pages folder) is a top-level React